Lesson Notes

MODULE 2 | LESSON 1

RESAMPLING EFFICIENT FRONTIERS

Reading Time	90 minutes
Prior Knowledge	Mathematics of classical portfolio theory and certain statistical knowledge
Keywords	Resampling, resampling efficient frontiers, Michaud's optimization, Morninstar's EnCorr, bootstrap techniques.

This lesson will describe the methodology of resampling the efficient frontier to obtain more robust estimates that do not suffer the same drawbacks as when practitioners rely on the simple optimization technique described in Lesson 4 of Module 1.

1. General Mean-Variance Optimization Problem

In Lesson 4 of Module 1, we learned several things about the efficient frontier. However, most of what we did was find the best portfolio for us given a required rate of return, which is the very definition of efficient frontier. There is no arguing that this is relevant. Still, if we recall the way we got to the formulation of the efficient frontier starting from an individual's utility function, i.e.,

$$U(w() = U(\bar{w}) + U'(\bar{w})(w - \bar{w}) + \frac{1}{2}U''(\bar{w})(w - \bar{w})^2 + R_3,$$

we concluded that individuals have a preference for expected wealth,$\bar{w}$, ($U(\cdot)$ is an increasing function of its argument), and also, individuals who are risk-averse will have an aversion to risk ($U''(\cdot)$ has negative sign). This leads us to write the following maximization program:

$$\begin{aligned}&\max_{w} w^T r - \frac{1}{2} w^T \Sigma w\\ &\text{s.t. } w^t \iota = 1,\end{aligned}$$

where the symbols have the same meaning as before. This is the standard program, but we could add additional constraints like maximum exposure to any single stock, minimum exposure, etc. We are also aware that, even if agents are risk-averse, it is quite possible that not everyone has the same risk-aversion to volatility. We could modify the QP above to read as follows:

$$\max_w w^T r - \frac{\alpha}{2} w^T \Sigma w$$
$$\text{s.t. } w^t \iota = 1,$$

where $\alpha \geq 0$ measures the intensity of risk-aversion.$\alpha = 0$ would describe an individual who has no consideration for risk,$\alpha = 1$ would describe our representative risk-averse agent,$\alpha = 2$ an investor who feels a much stronger dislike for risk than someone with $\alpha = 1,$ and so on.

Using the same portfolio of seven stocks as before and the same reference period, we find that for a traditional risk-averse individual $(\alpha = 1)$:

```
optimal weights:
0.7652182 1.7812181 2.0609607 -0.3036158
-1.2216889 0.6933467 -2.7754390
return achieved on portfolio:
1.175
volatility of optimal portfolio:
1.0166
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On the other hand, if the individual had much less appetite for risk (say the individual is near retirement and does not have enough years to make up for losses in their portfolio) and applies a larger penalty to risk, say $\alpha = 2,$ then we find:

```
optimal weights:
0.44033322 0.92022890 1.09255015 -0.04042926
-0.58977574 0.34565857 -1.16856583
return achieved on portfolio:
0.6811266
volatility of optimal portfolio:
0.5410759
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This is all interesting and one could toy with it by changing the value of the penalty factor α and investigate what happens to the solution.

In fact, it would be a great exercise for students who would like to master working with matrices (an important mathematical tool, in general) to solve this problem analytically instead of relying on numeric methods. The solution for the weights turns out to be:

$$w_{opt} = \frac{\Sigma^{-1}\iota}{\iota^T \Sigma^{-1} \iota} + \frac{1}{\alpha}\left(\Sigma^{-1} - \frac{\Sigma^{-1}\iota\iota^T\Sigma^{-1}}{\iota' \Sigma^{-1} \iota}\right) r.$$

Immediately, we notice that if $\alpha \to \infty,$ we get the weights for the minimum variance portfolio! That is what we would expect: If someone cannot stand any bit more variance than necessary, the global minimum variance portfolio is the only optimal portfolio for them. Hence, the solution we found makes sense.

2. Parameter Estimation and Issues

The reason why we considered the analytical solution is not just for the sake of doing some mathematics. The real motivation rests with the fact that the formula shows how the optimal portfolio (and similarly the efficient frontier) depends heavily on inputs like the covariance matrix, Σ, and the vector of expected return, r.

In practice, the parameters necessary to derive the efficient frontier are not known with certainty and, anyway, are bound to change over time. As we rely on estimates based on historical returns to estimate these parameters, the solution of the optimization problem is subject to instability in the sense that even small changes in the inputs could produce large differences in the optimal portfolios that we computed. In other words, noisy inputs lead to unstable portfolios.

In particular, it so happens that when a few assets have much higher average returns than others, a portfolio will tend to concentrate on those assets and this leads to portfolios that are not well-diversified. The issue is not so much the type of solution that we achieved and which is correct based on the inputs, but the fact that if the economic scenario changes, we will not reap the benefits of diversification.

Also, noisy estimates, changing a lot over even relatively short periods of time, lead to portfolios that require frequent re-balancing and which can be expensive to implement.

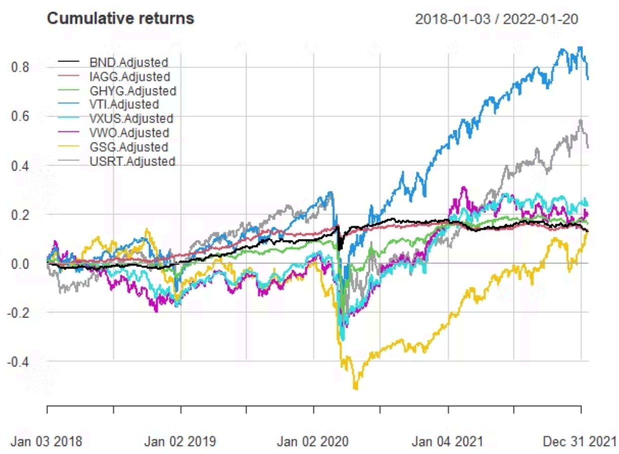
The COVID-19 crisis illustrates the instability we refer to. In order to make the example most useful, we will not use just U.S. equities but will instead look for different classes of assets to try being encompassing of a large variety of assets. The assets/classes of investments we will use are the following:

- the Vanguard Total Stock Market Index (VTI);
- the Vanguard FTSE Emerging Markets Index Fund ETF Shares (VWO);
- the iShares Global High Yield Corporate Bond (GHYG);
- the Vanguard Total Bond Market Index Fund ETF Shares (BND);
- the iShares Core International Aggregate Bond ETF (IAGG)
- the Vanguard Total International Stock ETF (VXUS)
- the iShares S&P GSCI Commodity-Indexed Trust (GSG)
- the iShares Core U.S. REIT ETF (USRT)

We consider the period between 2018 and the current date so that we have roughly two years before and after the start of COVID-19 crisis.

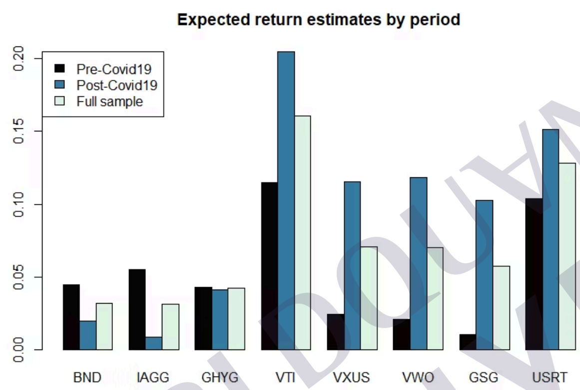
The plot of the cumulative returns is very clear in showing how much the behavior of the assets has changed over the two periods before and after COVID 19.

Figure 1: Cumulative Returns for Asset Prices for the Period Indicated



Let's look at the expected returns and the volatility over the two sub-periods and the entire sample, which is done in Figure 2.

Figure 2: Expected Returns Pre-COVID 19, Post-COVID 19 (Full Sample Period)



The bar chart for volatility shows a similar pattern of variation over the two subperiods (Figure 3). And it is likely that the correlations have changed as well to a certain extent.

Figure 3: Volatility Pre-COVID 19, Post-COVID 19, and over the Full Sample Period

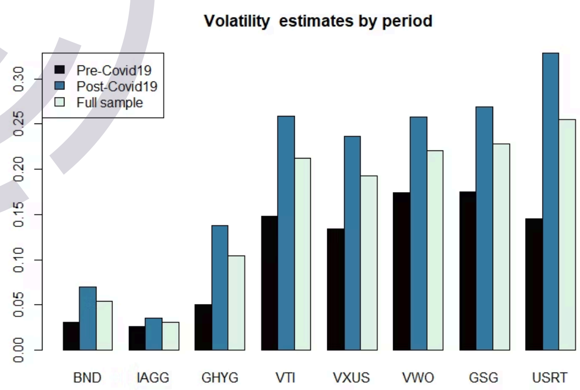


Figure 4 shows how the efficient frontiers are different among the different periods. This clearly leads to different weights of the individual assets in a portfolio and may produce high concentrations on fewer assets and force substantial and costly re-balancing. Also, notice that for during the COVID-19 period, the tangency portfolio is found much further to the right due to the much higher volatilities and the higher expected returns.

Figure 4: Efficient Frontiers Pre-COVID 19, Post-COVID 19, and for the Full Sample Period (The dots on each curve represent the tangency portfolio)

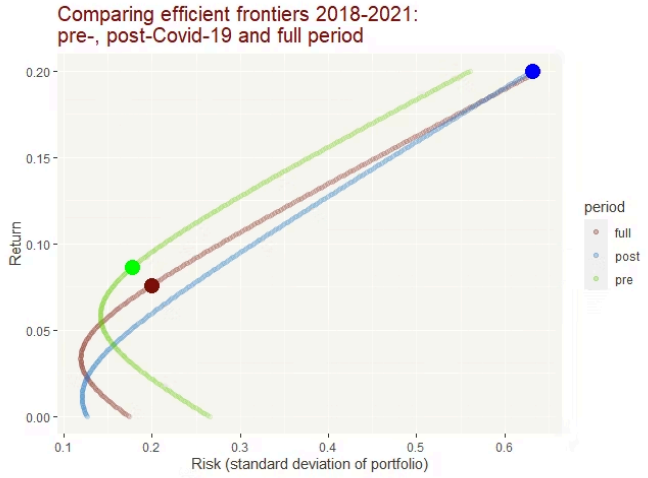
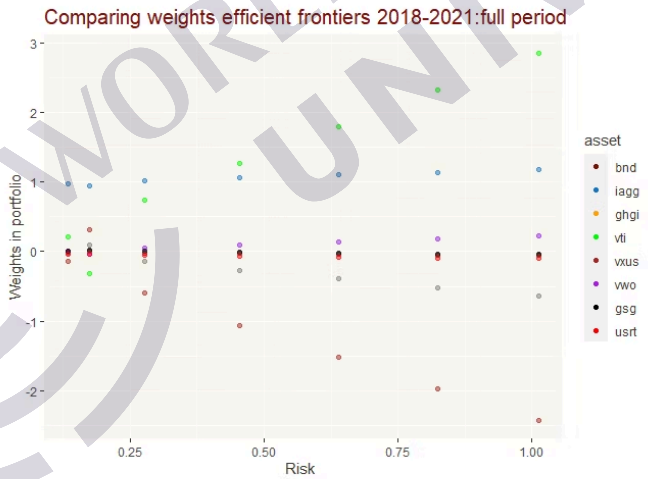


Figure 5 looks at the weights for the optimal portfolios based on specific amount of risks. This plot shows two important features:

- the portfolio tends to consist of just a few of the 8 assets (VTI, IAGG, VXUS) and as the desired return increases (and thus does the risk/volatility of the optimal portfolio), and
- a long position in VTI and a short position in VXUS tend to dominate all other weights more and more.

We looked at the results for the full period, but those for the other two subperiods are similar.

Figure 5: Weights of Assets for Portfolios on the Efficient Frontier



There is a relatively plausible explanation for the fact that the weights of some of the assets become bigger and bigger in proportion to those of the other assets. As the algorithm tends to maximize Sharpe's ratio and short selling is allowed, it achieves this by aggressively selling short stocks with the lowest estimated mean return and buying stocks with the highest estimated mean returns. Adding constraints to our optimization problem would alleviate the issue to some extent, but it would not change the fact that we are relying on estimated parameters and that even small differences in the estimates can produce large effects on the results.

3. Resampling the Efficient Frontiers

One solution to the problem of concentrated positions is to add constraints to the optimization problems. For instance, we could impose that, under no circumstances, the weights of all or some assets may exceed a certain % of the total portfolio. Similarly, we could impose restriction about the amount of short-selling that is allowed or even demand that there be no short-selling at all. Overall, these expedients will not allow any asset to become dominant or to be used as a source to generate funds (via short-selling).

Nevertheless, because the efficient frontier and the optimal portfolio rely heavily on the estimated mean returns, $\hat{\mu}$, and variance-covariance matrix, $\widehat{\Sigma}$ it is quite possible that our estimates are off by quite a bit vs. the true (and unknown!) value of these parameters, μ and Σ, respectively. We should reflect on the fact that we only see one sample out of an entire population of possible returns for our assets during the reference period. In particular, mean returns are very prone to estimation errors.

R. Michaud in 1998 began to pioneer the use of resampling techniques dealing with the problem of global asset allocations. In fact, Michaud has built a business around resampling (and as a matter of fact, the procedure was patented in 1999, U.S. Patent #6,003,018). His techniques have a name as well, **Michaud optimization or resampled efficiency.** Still, he is not the only one to have thought about this since Morningstar also has its own resampling technique, which is incorporated in their **EnCorr** software (Moringstar EnCorr User Guide). The two methodologies use different approaches and interested readers can find a comparison between the two techniques in the following paper: Morningstar vs. Michaud Optimization.

Michaud's procedure produces multiple sets of statistically equivalent (i.e. coming from a common underlying distribution) risk-return estimates around the original estimates. When the terminology ""statistically equivalent"" is used, it means that they all come from the same population/distribution. These sets of statistically equivalent returns allow us to compute many efficient frontiers, each optimized with respect to the particular sample we got from our assumed distribution for the inputs. In fact, each frontier will be different from the others, some maybe even very different from one another. The Michaud method calculates its efficient frontier by averaging all the frontiers that were produced using simulations. His method works like this:

Step 1. Given the time series of returns, estimate the expected returns of the m assets, which we call $\hat{\mu}$ and the variance-covariance matrix $\widehat{\Sigma}$ These can be estimated using MLE or any other statistical techniques (some may produce better estimates than others).

Step 2. With $\hat{\mu}$ and $\widehat{\Sigma}$ estimated earlier, we can determine the minimum variance portfolio and the maximum return portfolio using the suitable formulae

$$w = argmin_{w \in C} w^T \widehat{\Sigma} w$$

where C is the set of constraints for our problem, and

$$w = argmax_{w \in C} w^T \hat{\mu}$$

We call the return of the portfolio associated with these two systems of weights L and H, respectively. Note that if we set limits to the portfolio exposure to each assets, the maximum return will be finite

Step 3. Divide the range $L, H]$ in say $N-1$ subintervals (the reason for $N-1$ is that we want to have N points). This yields the following expected returns:

$$L \equiv E[r_{GMV}], r_2 \equiv E[r_{GMV}] + \delta, \ldots, r_N = E[r_{GMV}] + (N-1)\delta = H$$

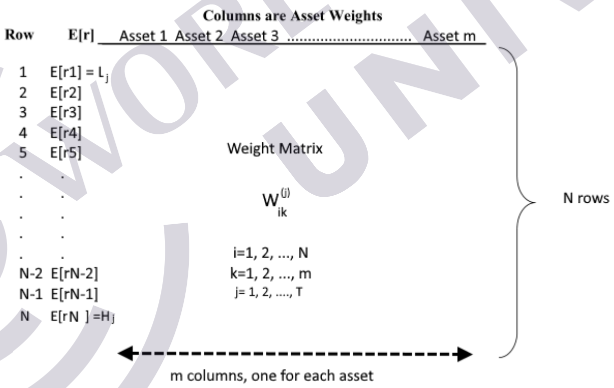
with

$$\delta = \frac{H-L}{N-1}.$$

Step 4. This is were the simulations begin. We make an assumption that the returns are distributed as a multivariate normal with mean $\hat{\mu}$ and covariance matrix $\widehat{\Sigma}.$We extract T random samples from this distribution. (Note that because we are sampling from a multivariate normal with covariance $\widehat{\Sigma}$, the returns we are sampling are not uncorrelated in general and reproduce the dependency that we see in the real markets). In other words, we are generating T samples of m returns, one for each of our assets from a population $N(\hat{\mu}, \widehat{\Sigma}).$.

Step 5. For each of the generated T samples, we estimate a new mean expected return, say $\tilde{\mu}_j$ and a new covariance matrix, say $\tilde{Sigma}_j$ where $j = 1, 2, \ldots, T.$ Then, we use these estimates to compute an efficient frontier, say EF_j using the N return points in the interval $[L, H]$. At this point, for each j, we should have a matrix of weights $W_{ik}^{(j)}$ like in the picture immediately below. Note that at this stage it is better to replace L and H with L_j and H_j, but the process works exactly as described in Step 2.

Figure 6: Resampling Weights using Michaud's Efficient Resample Technique

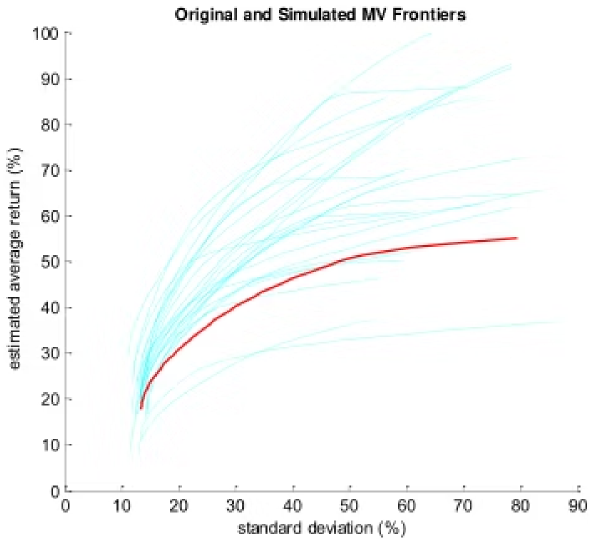


Step 6. We have now T matrices of weights as described in the picture above. That means that for every entry W_{ik} we have T values, $W_{ik}^{(j)}, j = 1, 2, \ldots, T,$ one value for each simulation. Then we average each element of the matrix over to get $W_{ik}^* = \frac{1}{T}\sum_{j=1}^{T} W_{ik}^{(j)}.$

Step 7. With the new weights W_{ik}^* we can redraw the efficient frontier by using the original means, $\hat{\mu}$ and variance-covariance matrix $\widehat{\Sigma}.$ This yields the resampled efficient frontier.

What we get after all this looks like the picture immediately in Figure 7.

Figure 7: The Original Efficient Frontier (red) and the Resampled Frontiers (light blue)



Source: Michaud, Richard & Michaud, Robert. (2007). Estimation Error and Portfolio Optimization: A Resampling Solution. Journal of Investment Management. Vol. 6. pp. 8 - 28. Image reproduced with permission by the authors.

Resampling efficient frontiers with a large number of portfolios and implementing it properly to guarantee convergence requires a very heavy computational burden. At the same time, this is one of those problems that can benefit from parallel programming.

Once we determine an efficient frontier using Michaud's resampling methodology, if we notice that the efficient frontier obtained from the original data is sufficiently close (say in a 95% confidence level region), we can use our result with a certain degree of confidence. However, if we notice that the efficient frontier is quite far apart from Michaud's, we cannot simply attribute the differences to statistical factors, and we must investigate the problem further.

Also, we have proposed a multivariate normal as our underlying distribution, but we could have used a multivariate t distribution or other distributions, in principle whenever we have a reason to decide that one family of distributions is better than the others.

Overall, estimating ρ and Σ using sample means is not considered the best possible approach and other estimating procedures are possible. One such method is **shrinkage estimation**. Let's say that ρ_i is the mean for asset i(out of m assets), and we define $\tilde{\rho} = \frac{1}{m}\sum_{i=1}^{m}\rho_i,$ i.e., it is the mean of means (Statisticians refer to it as the "grand mean").

The shrinkage estimator for the mean is simple:

$$\tilde{\rho}_i = \gamma\rho_i + (1-\gamma)\tilde{\rho}.$$

This procedure is implemented to bring the means closer to each other. One may wonder why. The rationale is to reduce the variance of the estimator. Can it be done at no cost? No, there is a price to pay and the tradeoff is that using a shrinkage methodology introduces bias (unless the true means for the returns happen to be all the same, an unlikely event).γ is the shrinkage parameter, and it is a number between 0 and $1.$ If $\gamma = 0$ all $\tilde{\rho}_i = \tilde{\rho},$ the grand mean and the variance is minimal. However, in that case, the bias is large. Shrinkage estimators, we stress, are only one approach and do not always produce dramatic improvements, but they do work in many cases.

There is a shrinkage estimator for the covariance matrix as well, but that is more complex than for the case of the mean. One approach suggests to write the covariance matrix as $\Sigma = M\Lambda M^T$

where M is the matrix of normalized eigenvectors and Λ is the matrix of the eigenvalues of Σ. Note, that this can be done because of specific theorems in linear algebra.

Then, one "shrinks" the eigenvalues of the matrix Λ. For instance:

$$\tilde{\lambda}_i = n\lambda_i/\alpha_i$$

and

$$\alpha_i = n - p + 2\lambda_i \sum_{j \neq i} \frac{1}{\lambda_j - \lambda_i}$$

But, this approach (known as the *Stein estimator*) does not work all the time because it can produce negative eigenvalues, which should not happen when a matrix is positive definite like a covariance matrix. There are others, but this is not the place for a sophisticated investigation of the topic. Students interested in this topic should consult B. Efron and T. Hastie's Computer-Aided Statistical Inference, Chapter 7, Section 1, pp. 91-94.

In general, the means are the critical input. The variance/covariance matrix is important too, but while even small changes in the estimated means can have a large impact in the determination of the optimal portfolio weights, this is not so for the variance/covariance matrix.

The relevant takeaway is that when working with small samples, a shrinkage estimator may help improve the results obtained from resampling procedures.

Michaud's technique has been tested by practitioners and academics over the years. For instance, in 2003, Markowitz and Usmen compared it to the Bayesian portfolio optimization method, and while Markowitz and Usmen expected the Bayesian strategy to do better than the resampling approach, surprisingly, Michaud's strategy did better. But not everyone agrees that this conclusion holds all the time. Students who are interested in this topic should read A Theoretical Foundation of Portfolio Resampling.

Morningstar's EnCorr uses resampling as well. However, it uses a different approach. Morningstar's algorithm divides the risk/volatility spectrum spanned by the Markowitz efficient frontier into equal lengths of bins of standard deviations. The minimum bin starts at the minimum variance portfolio and the maximum bin ends at the maximum return portfolio. EFs are simulated using a volatility within its bin. The resulting Morningstar efficient frontier is computed as the average of the portfolios in each bin and displayed as the risk and return of the average portfolio.

Note: Michaud's portfolio resampling methodologies is a patented procedure. This means nobody can use or distribute the algorithm without licensing it or obtaining permission. This is important for a quantitative analyst as much as for a portfolio manager since the use of the methodology without licensing could produce monetary fines for the professional or the firm. It would also reflect badly in their business relations with clients and may be in breach of the fiduciary duty to clients.

Students interested in understanding more in depth the nuances of portfolio resampling, and not just Michaud's, are encouraged to read the paper by Frahm listed among the references.

4. Accuracy of Estimates of the Sharpe Ratio for the Tangency Portfolio

Assume that we want to find the tangency portfolio that maximizes the Sharpe ratio. This is something that we know how to do very well by now using our returns and the estimated parameters (mean returns and covariance matrix). Clearly, as this is just one sample of returns over the reference period out of many possible, we would like to check whether the Sharpe ratio of the portfolio we find is an over- or an under-estimation of the actual one.

The bootstrap technique is very useful to answer this type of question. In this case, we treat the sample we have as if it were not a sample, but the actual population of stock returns. Consequently, the mean returns and covariance matrix are no longer estimates but true parameters. In other words, we make the sample available to be our population and the Sharpe ratio for the tangency portfolio from this sample population becomes the true Sharpe ratio.

Now we employ a resampling scheme via bootstrap and not by simulating new data. The process is rather simple and works like this:

Step 1. Draw M samples from the data (resamples).

Step 2. Decide on the resample size $n < N$, with N the number of returns for each asset in our portfolio. Deciding what n to use is a practitioner decision. Sometimes, one tries the experiment using different values of n.

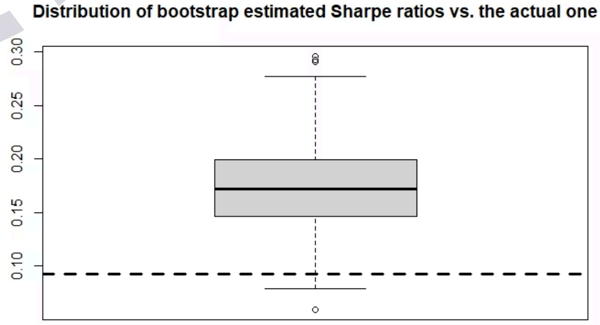
Step 3. For each resample, we estimate a new vector of mean returns and a covariance matrix. With these, we can compute the optimal (Sharpe) portfolio. This will give us M optimal portfolios. Overall, we will have M systems of weights and M Sharpe ratios, one for each of the bootstrap samples.

Step 4. Take the mean of the weights for the M optimal portfolios.

Step 5. The re-sampled efficient frontier is then found by multiplying the average weights from Step 4 by the actual returns and the risk by using the covariance formula for the portfolio and the same weights from Step 4.

The results of our simulations are shown in Figure 8. The bootstrap experiment shows that our process to identify the efficient frontier tends to overestimate the actual one, that is things look better than they really are. The data used for the simulation is obtained from the pre-covid period without short sale restrictions.

Figure 8: Distribution of Sharpe Ratios vs. Actual Ones using Bootstrap Resampling



The overestimation of the efficient frontier and the Sharpe ratio produces higher confidence in the process than it probably should. Can the issue be alleviated or reduced? If we introduced short-sale restrictions, one could see that the bootstrap estimates fall much closer to the actual Sharpe ratio.

Statistical remark about bootstrapping with time series: This seems relatively simple and, to a certain extent, it is. However, we should stress the fact that we are bootstrapping multivariate data and, in particular, time series. When resampling multivariate time series data, the dependencies within the observation vectors must be preserved. In other words, we cannot just adopt a re-sampling scheme with replacement because that would destroy dependency. So how do we do it? Thankfully, it is not too complicated. We assumed earlier that we had samples of N data (i.e., we know the returns for all our assets during a period of N days, months, or whatever the reference period is. Then, we identify our sample rows with a number: 1, 2, ..., N. When we take the M resamples, we resample the number of the row and then we take the returns for all assets along that particular row.

5. Conclusion

We derived the general mean-variance optimization solution and noticed that it relies heavily on the estimated mean vector and covariance matrix for assets' returns over the reference period. When these returns are non-stationary, the algorithm produces solutions that can be very volatile and concentrated. To mitigate these issues, we introduced Michaud's optimization, or resampled efficiency, and learned how to derive a better efficient frontier, which takes into consideration that mean returns and covariance matrices are not parameters but simply estimates. In the last section, using bootstrap, we saw that the traditional method of deriving optimal portfolios overestimates the efficient frontier and thus predicts better results than one would actually achieve.

We stress here, once again, that Michaud's methods cannot be used without license or permission because it is patented. The next lesson will broaden the problem of optimal portfolio selection by adding skewness, kurtosis, and liquidity to the problem.

References

- Frahm, G. "A Theoretical Foundation of Portfolio Resampling." *Theory and Decision*, 2014, pp. 1–30, http://dx.doi.org/10.2139/ssrn.2085586.
- Michaud, Richard O., and Robert O. Michaud. "Estimation Error and Portfolio Optimization: A Resampling Solution." *The Journal of Investment Management*, vol. 6, no. 1, 2007, pp. 1–8.